

MacroWatch External References

Every External System, Tool & API — Why Selected, Role, Conditions of Use

v1.1-stable | May 8, 2026 | GitHub: [mlibrizzi/MacroWatch](#)

1. Anthropic Claude API

Why Selected

Claude provides the intelligence layer — transforming raw data into coherent macro narratives, regime classifications, and positioning signals. Selected over GPT-4 for stronger analytical reasoning on financial topics, more reliable structured JSON output, and better ability to follow complex system prompts without drift.

Role in MacroWatch

- **Intel Tab:** Generates full macro briefing — thesis, regime, alerts, key risks, watch list, Dalio lens, positioning signals — anchored to live market data
- **Weekly Tab:** Generates Fed policy narrative (Powell status text, QT description). Numerical values are hardcoded separately to prevent hallucination.
- **QTR Tab:** Generates government report summaries anchored to verified hardcoded values
- **Monthly Tab:** Generates basis trade narrative

Technical Implementation

Component	Details
Model	claude-haiku-4-5-20251001 — fastest and most cost-effective for structured JSON generation. Sufficient for macro analysis tasks.
Max tokens	4,000 (set in src/api.js). Intel briefing typically uses 3,000-4,000 tokens. Increased from 2,000 after Intel tab was silently truncating responses.
API Proxy	api/claude.js (Vercel serverless function). All Claude calls go through this server-side to protect the API key. Adds system prompt enforcement that strips markdown and forces raw JSON output. Prevents client-side key exposure.
Client Helper	src/api.js — callClaude(prompt, systemOverride). Falls back to {raw: text} if JSON parsing fails, which triggers "Briefing parse error" message instead of silent crash.
Key storage	VITE_ANTHROPIC_API_KEY in Vercel environment variables. Never committed to GitHub.

Cost & Limits

Item	Details
Pricing model	Per token. Input + output tokens billed separately. Haiku is significantly cheaper than Sonnet or Opus.
Approximate	~\$0.25/million input tokens, ~\$1.25/million output tokens (Haiku pricing — verify at

cost	anthropic.com/pricing)
Rate limits	Depends on tier. Paid tier recommended for production. Free tier has strict limits that cause timeout errors.
JSON reliability	Claude is instructed via system prompt to return ONLY raw JSON. Additional enforcement in api/claude.js strips any markdown fences that slip through.

2. GitHub

Why Selected

GitHub provides version control, code history, and the automatic deployment trigger for Vercel. Every push to the main branch triggers a Vercel deployment within seconds. GitHub also enables the iMac Terminal workflow using the REST API — all code changes can be made without a local git installation.

Role in MacroWatch

- Source code repository for all MacroWatch files
- Version control — every change tracked with commit message, SHA, and timestamp
- Deployment trigger — Vercel watches main branch and auto-deploys on every push
- Stable tags — v1.0-stable, v1.1-stable enable rollback to known good states
- Documentation storage — /docs folder contains all reference documents

GitHub Personal Access Token — Complete Workflow

MacroWatch uses GitHub's REST API directly from iMac Terminal via Python scripts. No git installation required. All changes flow through the heredoc pattern.

Step	Action	Details
1. Generate Token	Create PAT on GitHub	github.com → Settings → Developer settings → Personal access tokens → Tokens (classic) → Generate new token. Required scope: repo (full control). Set expiration as desired.
2. Export in Terminal	Set environment variable	<code>export GH_TOKEN='your_token_here'</code> — must run each new Terminal session. Not persisted between sessions by design (security).
3. Fetch Current File	GET request	urllib.request.urlopen reads current file content from raw.githubusercontent.com. Also fetches SHA needed for update.
4. Modify Content	Python string replace	Surgical: <code>code.replace(exact_old_string, new_string)</code> . Always verify change applied with print confirmation. Check smart quotes = 0.
5. Upload via API	PUT request	Base64 encode modified content. PUT to <code>api.github.com/repos/mlibrizzi/MacroWatch/contents/path</code> with sha of current file. Commit message required.
6. Vercel deploys	Automatic	GitHub push triggers Vercel build. Typical time: 15-20 seconds. Check Vercel dashboard for green status before testing.
7. Test	curl or	Always test via curl API endpoint or browser after deploy. Never declare success without

	browser	verification.
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Common Issues:

- **401 Unauthorized** = expired token. Re-generate at github.com and re-export.
- **"NOT FOUND" in Python script** = old string doesn't match exactly. Fetch file first and print the exact string before replacing.
- **Smart quotes (chr 8216/8217/8220/8221)** = will break JavaScript build. Always check count = 0 before uploading.
- **Heredoc prompt (>)** = stuck in heredoc. Press Ctrl+C to exit.

3. Vercel

Why Selected

Vercel provides zero-configuration deployment for React/Vite apps with built-in serverless functions. The free hobby tier is sufficient for MacroWatch. Vercel's global edge network ensures fast load times, and automatic HTTPS enables PWA installation on iPhone home screen.

Role in MacroWatch

- Hosts the static React/Vite bundle (frontend app)
- Runs `/api/*.js` as serverless functions — all external API calls happen server-side
- Stores API keys securely as encrypted environment variables
- CDN edge caching for fast global delivery
- Automatic HTTPS — required for PWA installation on iPhone
- Auto-deploy from GitHub main branch — push code, app updates automatically

Key Configuration

Setting	Value	Notes
Plan	Hobby (free)	Sufficient for personal use. No SLA. Function timeout: 10 seconds.
Live URL	macro-watch-phi.vercel.app	Primary domain. Also accessible via deployment-specific URLs.
Build command	npm run build	Auto-detected. Runs vite build → outputs to /dist
Node.js version	24.x	Set in Deployment Settings → Runtime Settings
Function region	iad1 (US East)	Closest to Treasury API and FRED servers
Cache purge	CDN → Caches → All content → Purge	Use when stale data persists despite code changes
Environment variables	Settings → Environment Variables	5 API keys stored. Click eye icon to reveal (sensitive keys require confirmation).

4. Active External APIs

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API	URL	Cost	Limit	Key	Data & Notes
FRED St. Louis Fed	api.stlouisfed.org	FREE	120 req/min	VITE_FRED_API_KEY	Most comprehensive financial data Provides: Treasury yields (DGS2/DGS10/DGS30), Inflation (CPIAUCSL/CPILFESL), GDP (A191RL1Q225S), Industrial Production (WALCL/TREAST/WSH), TGA (WDTGAL), VIX (VIXCL), Oil (DCOILWTICO), Brent (DCOILBRETEU), Labor (UNRATE/PAYEMS/IC:), and many more daily series. Official govt data.
Finnhub	finnhub.io	FREE tier	60 req/min	VITE_FINNHUB_API_KEY	Real-time US stock quotes for AAPL/MSFT/NVDA/GOOGL/AMZN/ETF quotes for index funds like SPY/QQQ (×29 = NDX), Dividends, earnings: EPS actual vs expected for companies. Note: free tier has a 5-min delay for VIX (^VIX returns null).
gold-api.com	gold-api.com	FREE tier	Limited daily	VITE_GOLD_API_KEY	Real-time gold (XAU/USD) spot price: XAU/USD. Used in api/metadata.
OilPriceAPI	api.oilpriceapi.com	FREE tier	1,000 req/month	VITE_OILPRICE_API_KEY	Near real-time WTI (WTI) / BRENT (BRENT_CRUDE_USD) on free tier. 24h change in price. Investing.com (WTI) / Selected May 8, 2026 caused MacroWatch to report on Iran deal news (Macroeconomic Advisers).
open.er-api.com	open.er-api.com	FREE	No stated limit	None required	EUR/USD, USD/JPY, GBP/USD on free tier. No API key required. api/markets.js for FX rates.
Treasury Fiscal Data	api.fiscaldata.treasury.gov	FREE public	No stated limit	None required	Official US Treasury. Auction data: security_term, auction_id, bid_to_cover_ratio, indicating primary_dealer_tender. Same-day publication of bidder data (BTC/india) published hours after. PENDING during this time. api/auctions.js.
Treasury TIC	ticdata.treasury.gov	FREE public	No stated limit	None required	Major Foreign Holders File: slt_table5.txt (table of US Treasury securities). Released ~45 days after monthly holdings for 2013-2026. api/tic.js. Feb 2026 data.

Anthropic API	api.anthropic.com	PAID per token	Tier-dependent	VITE_ANTHROPIC_API_KEY	AI narrative generation details. Model: claude Proxied via api/claude
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5. APIs Evaluated But Not Selected

API	Why Evaluated	Why Not Selected	Status
Twelve Data	Tested for DXY (~98), VIX. Signed up for free account.	DX-Y.NYB symbol not found. VIX not available. Only returned UUP/UDN ETFs for "dollar index" search. Free tier: 8 req/min — insufficient for multiple simultaneous calls.	REJECTED
Alpha Vantage	Considered for DXY index data. Claims to support market indices.	Only 25 requests/day on free tier — far too low for production dashboard that makes multiple calls per page load.	REJECTED
API Ninjas	Tested for WTI/Brent commodity prices. No credit card required.	Free tier only allows 7 <i>rotating</i> commodities per week — WTI and Brent may not be available in a given week. Unreliable for production use.	REJECTED
EODHD	Evaluated for real-time stocks, DXY, historical data at ~\$22/month. Has direct NDX quote, headline DXY.	Commodities API uses FRED data (same source we already have — no improvement for oil). DXY and historical data are valuable. Planned for Phase 2 when trend sparklines are built.	PLANNED Phase 2 ~\$22/mo
Yahoo Finance (yfinance)	DX-Y.NYB ticker for headline DXY. No API key required.	Not an official API — web scraping only. Fragile, subject to breaking without notice. Not suitable for production dashboard.	REJECTED
Bloomberg Terminal	True when-issued yield for auction tail calculation. Industry standard.	~\$25,000/year — not viable for personal dashboard. Used by institutional desks only.	REJECTED — cost
Interactive Brokers API	True auction tail (when-issued yield), futures curves (gold/Treasury/oil), USD/THB, USD/SGD, headline DXY.	Account opened May 3, 2026. Pending approval and minimum funding. Will be the primary Phase 3 data source. Essentially free with a funded trading account.	PENDING IB APPROVAL
OilPrice.com (direct scrape)	Real-time oil prices directly from oilprice.com website.	No official API — HTML scraping only. Fragile and against terms of service. OilPriceAPI.com is the official API wrapper for the same data source.	REJECTED

6. Known Data Gaps & Planned Solutions

Gap	Current Workaround	Impact	Planned Solution	Est. Cost
Headline DXY (~98)	Showing Fed broad trade-weighted index (~118) — different index, different methodology	Medium — users expect ICE DXY. Labeled clearly as "Fed Broad USD (not ICE DXY)"	EODHD subscription (Phase 2G)	~\$22/month

True Auction Tail	Showing N/A. Requires when-issued yield (pre-auction forward market price)	High — tail is the most important single auction stress metric. Key Dalio signal.	Interactive Brokers when-issued yield (Phase 3D)	~\$0 with funded IB account
Synthetic Auction Tail	Not yet calculated	Medium — acceptable approximation per Dallas Fed methodology: high yield minus prior day close yield	Treasury Fiscal Data high yield + FRED prior day yield (Phase 2, free)	\$0
Futures Curves	Not implemented	Medium — gold futures curve, Treasury futures, oil term structure missing	Interactive Brokers API (Phase 3D)	~\$0 with IB account
BTC 6-month rolling avg	Showing "avg undefined" — hidden in UI	Low — provides historical context for each auction metric	Treasury Fiscal Data API historical query (Phase 2, free)	\$0
5Y5Y Breakeven	Not shown	Medium — inflation expectations 5 years forward, key Fed policy input	FRED series T5YIFR (Phase 2F, free)	\$0
Corporate Credit Spreads	Not shown	Medium — HY spreads signal credit stress before equities react	FRED series BAMLH0A0HYM2 (Phase 2F, free)	\$0
Bitcoin Price	Not shown	Low-Medium — Dalio increasingly acknowledges Bitcoin as emerging hard money	CoinGecko free API or Finnhub (Phase 2F, free)	\$0
Real-time EUR/USD	Hourly update via open.er-api.com	Low — hourly delay acceptable for macro monitoring	EODHD real-time forex (Phase 2G)	~\$22/month
Interest Payments / GDP	Not shown	Medium — classic Dalio "plaque" indicator. Currently ~3.5% of GDP and rising.	Treasury + BEA data, hardcoded quarterly (Phase 2F)	\$0

7. Development Tools & Stack

Tool	Version	Role	Notes
React	18.x	Frontend UI framework	Component-based. All UI in single App.jsx for simplicity. Hooks: useState, useRef, useEffect.
Vite	5.x	Build tool and dev server	Fast HMR in development. Optimized production builds. Auto-detects React. Output to /dist.
Recharts	Latest	Chart library	Bar charts for auction demand trends (BTC, Indirect %, Dealer %). Pure React component library.
Python 3	3.9 (macOS)	Deployment scripts	All GitHub API calls via Python heredoc scripts in Terminal. Uses only standard library (urllib, json, base64, os). No pip installs needed for deployment.

Node.js	24.x	Vercel serverless runtime	Set in Vercel Deployment Settings. Serverless functions use ES module syntax (export default).
macOS Terminal (zsh)	Default	Development environment	All code changes made via Terminal. Heredoc pattern: <code>python3 << 'EOF' ... EOF</code> . Ctrl+C to exit stuck heredoc.

8. Security Practices

Practice	Implementation
API Key Protection	All 5 API keys stored as Vercel environment variables. Never committed to GitHub. Never exposed in frontend JavaScript bundle. All external API calls proxied through <code>/api/*.js</code> serverless functions.
GitHub Token	Personal Access Token exported as shell variable each session — not persisted. Minimum required scope (repo only). Regenerate when expired.
No User Data	MacroWatch collects no user data. No login, no tracking, no cookies. Pure read-only dashboard.
HTTPS Only	Vercel enforces HTTPS automatically. Required for PWA installation and service worker functionality.
Smart Quote Prevention	All Python deployment scripts check <code>sum(code.count(chr(c)) for c in [8216,8217,8220,8221]) == 0</code> before uploading. Smart quotes (from copy-paste) break JavaScript builds.